Protocol for the Development of an Instrument for Assessing Risk of Bias of Randomized Trials in the Context of Systematic Reviews

1. Background

Clinical researchers and methodologists generally agree that randomized controlled trials (RCTs) and systematic reviews of such trials provide the most trustworthy evidence regarding causal inferences and effects of interventions.[1] The effect estimates presented in RCTs can, however, be biased due to flaws in design and execution of the studies. Given this concern, risk of bias assessment of individual studies has become an essential part in systematic reviews which influences the decision of whether to rate down GRADE certainty of evidence due to risk of bias.[2]

Investigators have developed many instruments for assessing risk of bias of RCTs.[1 3-7] However, even the most widely used tools - those developed by the Cochrane Collaboration[1 4] - suffer from limitations. Limitations of the first Cochrane risk of bias tool[1] include the "unclear" response option that fails to take advantage of reasonable inferences regarding presence or absence of risk of bias,[8 9] and the confusion about rating domains of incomplete outcome data and selective reporting.[10] Aiming to overcome these and other limitations, the Cochrane group developed the revised Cochrane instrument (RoB 2).[4] Although sophisticated, the RoB 2 has proved to have low inter-rater reliability, complexity of implementation, and challenges in its application.[11]

Therefore, we aim to develop a new user-friendly and credible instrument for assessing risk of bias of RCTs in the context of systematic reviews.

2. Ground rules of the instrument

The operations committee developed the following ground rules:

- The instrument aims to assess risk of bias of RCTs in the context of systematic reviews.

- The objective is to develop a user-friendly instrument: item presentation will be simple and straightforward; making judgements not overly complex or difficult.
- We define bias as a systematic error or systematic deviation from the truth.
- We assume that systematic reviewers will use the GRADE approach to assess certainty of evidence.
- Decisions should be consistent with the GRADE system in distinguishing risk of bias from imprecision (random error), indirectness (applicability), and publication bias. Reporting quality represents another issue to distinguish from risk of bias.
- The instrument currently addresses only risk of bias assessment of individually-randomized parallel-group trials. We leave the risk of bias assessment of cluster-trials and crossover trials for future consideration.
- This instrument will not include items that address the detection of fraud.

3. Methods

3.1. ASSEMBLE GROUP

3.1.1. Operations committee

The operations committee includes 5 members (Gordon Guyatt, Romina Brignardello-Petersen, Reed A Siemieniuk, Dena Zeraatkar, and Ying Wang). The operations committee came up with the idea of developing the new instrument and is responsible for execution of review work, organizing materials, and presenting proposals to the panel.

3.1.2. Panel

To complement the operations committee, we will invite additional experts in risk of bias methodology and evidence-based medicine education to join the panel.

Operation committee members invited Matthias Briel and Paul Glasziou to join as methodological experts because they could have substantial contribution. Operation

committee members suggested Sheri Keitz to join because she is an exceptionally astute evidence-based medicine educator.

We will invite 10 other risk of bias methodological experts and 2 other experts in evidence-based medicine education, which we will randomly select from a list of risk of bias methodological experts and a list of internationally recognized experts in evidence-based medicine education, stratified by gender and region.

Risk of bias methodological experts

We will collect potential risk of bias methodological experts from a list of authors of risk of bias methodological papers.

We will identify risk of bias methodological papers from four resources:

a) References of existing RCT risk of bias instruments;
b) References of RCT risk of bias instrument guidance documents describing the use of RCT risk of bias instruments;
c) Eligible papers that are suggested by operations committee members;
d) References of eligible papers that are identified from the above three resources.

Papers in which authors stated explicitly or implicitly indicated that what they addressed in the paper is a risk of bias issue, are eligible. Table 1 presents possible classification of eligible papers.

Eligible risk of bias methodological experts should be first or last or corresponding author of at least one eligible paper and co-author of at least two other eligible papers.

Evidence-based medicine educators

The operations committee members will suggest a list of internationally recognized experts in evidence-based medicine education from which we will randomly select two.

3.2. DEVELOPMENT OF THE INSTRUMENT

Development of the new RCT risk of bias instrument will consist of 4 steps: collection of potential candidate items for the instrument and develop an organized item list; collection of empirical evidence; a panel process to identify items that will be included in the new instrument and design the instrument; and user testing of the new instrument.

Step 1 - Collect potential candidate items and develop an organized item list

We will collect potential candidate items from three resources: included items of the existing RCT risk of bias instruments; items regarded as "other bias" in systematic reviews; comments on Cochrane risk of bias tool 1.0.

a) *We will conduct a systematic survey of existing RCT risk of bias instruments published in the last decade*

This systematic survey will include instruments that purport to assess risk of bias, internal validity, quality, or methodological quality of RCTs. Typical types of the instruments include scales, checklists, and domain-based tools. We will only include instruments published (or presented in websites) in 2010 or later.

We will search Medline, Embase, Web of Science, and Scopus from 2010. We will develop the search strategy in collaboration with an experienced research librarian. We will scan the reference lists of existing systematic surveys of similar topics to identify other potential eligible studies. We will extract included items of the existing instruments.

b) *Collect the items regarded as "other bias" in systematic reviews from an existing systematic survey*

Cochrane risk of bias tool 1.0 has a domain of “other bias”. A systematic survey published in 2019 summarized the items that Cochrane reviews regarded as “other bias” when they assessing risk of bias using Cochrane risk of bias tool 1.0.[12] We will

extract items from this systematic survey.

c) *Collect items from an existing systematic survey of published comments of Cochrane risk of bias tool 1.0*

A review published in 2016 summarized the comments of Cochrane risk of bias tool 1.0. Some of the comments were related to the additional items that may need to be included in the RCT risk of bias instrument.[13] We will extract items from this systematic survey.

After collecting items, we will conduct a survey of panel members regarding what issue each item addresses: risk of bias, applicability (imprecision), random error (imprecision), reporting quality, or none of the above. Based on the number of respondents that choose each option, we will classify items into three categories:

Category 1: items that the majority thought address risk of bias

Category 2: items that the majority thought do not address risk of bias

Category 3: items with substantial disagreement regarding whether they address risk of bias

Step 2 – Collect empirical evidence

Meta-epidemiological studies provide empirical evidence regarding how risk of bias issues influence effect estimates. We will conduct a systematic survey of meta-epidemiological studies examining the influence of potential risk of bias issues (items in category 1 and 3) on effect estimates in RCTs.

Three existing systematic surveys on this topic searched till to 2015.[14-16] We will review meta-epidemiological studies included in these three systematic surveys and assess their eligibility. We will conduct an updated search of Medline, Embase, Web of Science, and the Cochrane Database of Systematic Reviews from 2015.

Step 3 – Panel process

We will hold panel meetings to discuss and decide which items to be included in the new instrument; response options for single items, and to develop a manual that helps systematic reviewers completing the instrument.

We will use the below item selection criteria to help deciding which items to be included in the instrument. All criteria are desirable but not essential, i.e., final decision is up to the panel.

- Clearly risk of bias issue rather than imprecision, indirectness, publication bias, or reporting quality
- Theoretical or logical argument for why the item is important
- Information required to make judgement on the item is commonly reported in trials
- Non-expert systematic reviewers can make the judgment easily
- Problem occurs more often than rarely
- Empirical evidence supports item influence on effect estimates

The panel will first discuss items in category 2 (items that the majority thought not address risk of bias), then category 1 (items that the majority thought address risk of bias), and finally category 3 (items with substantial disagreement regarding whether they address risk of bias).

Since some items are highly related and only differ in details, during the item selection process, items may be first considered by groups if necessary. The panel will discuss the detailed issues during item selection or when developing instructions for items.

Step 4 – User testing

a) User testing with junior systematic reviewers

We will conduct a user testing of the new instrument to identify challenges experienced by potential users in applying the instrument, which will be used to improve the instrument. We will include people who have assessed risk of bias of RCTs

for at least one systematic review and have never led any systematic review of RCTs themselves. We plan to first recruit 15 people. To achieve saturation, we will continue to enroll people until they do not identify any new issues or problems in applying the new instrument.

Our panel members will send an invitation email to the potential eligible participants that they know. In the email, panel members will introduce this study, confirm the potential participants' eligibility, and ask their interests in participating in this study. The consent form will be sent with the email. If potential participants' eligibility is determined and they agree to participate, they will sign the consent form and send it back to the panel.

After obtaining consent, we will email participants a link to a LimeSurvey survey to collect their demographic information (Box 1).

Box 1. Demographic survey

- Participant ID
- Gender
- Country
- Clinical background: physician vs. pharmacist vs. nurse vs. others (specify) vs. no clinical background
- Student status: undergraduate student vs. master student vs. PhD student vs. not a student
- How many systematic reviews have you assessed risk of bias of included RCTs for: 1-2 vs. 3-5 vs. >5

Based on the demographic information, we will select 15 people that vary with respect to gender, country, clinical background, student status, and number of systematic reviews in which they have assessed risk of bias of RCTs.

We will select trials based on the following criteria:

i) Trials must be individually randomized parallel-group trials (described as

"randomized" by trial authors).

ii) We will prioritize trials with which systematic reviewers will face difficulty in assessing as many items in the instrument as possible (for step 1 we considered the trials in which judgements were probably yes or probably no as challenging; for step 2 we considered it challenging when the assessment involved judgement regarding whether the failure of implementation of methodological safeguard resulted in risk of bias). Challenge in each item must be faced in at least one final selected trial.

iii) Risk of bias assessments for the final selected trials vary across the response options for each item.

Panel members will suggest potential trials. To identify trials that meet our criteria, Ying Wang and Gordon Guyatt will apply the risk of bias instrument to the trials that the team members suggest. We plan to first select 5 trials. If 5 trials prove insufficient to meet our criteria, we will select more trials until we find sufficient trials that raise issues with each item. Ying will discuss trial selection with Gordon Guyatt and possibly with other panel members.

We will conduct a think-aloud interview with each of the 15 people online via Zoom. Prior to the interviews, we will send one trial and draft of the instrument and the manual to each participant. During the interview, participants will apply the instrument to the assigned trial. As they apply the instrument, we will instruct them to read the question, instruction, and manual for each item aloud and verbalize what they are thinking as they do so. When they apply each item to the trial, they will verbalize all the thought processes that lead to their assessment. The aim is to understand any challenges that participants have in comprehending and applying items to the trials. After they finish applying the instrument, we will ask their overall experience. We will record and transcribe the interviews, and extract participants' challenges in comprehending and applying the instrument.

As soon as we find a clear problem in the instrument or manual, we will make a correction immediately and continue the use testing with the corrected version.

b) User testing with systematic review experts

We will conduct another user testing of the instrument with systematic review experts to identify concerns or questions systematic review leaders may have about the instrument and the manual, which will be used to improve the instrument.

We will search Cochrane Library, randomly select Cochrane systematic reviews and identify the first or last or corresponding authors. If the authors have led at least five systematic reviews of RCTs, by checking in PubMed and Google, we will invite them to participate in the user testing. We will first recruit 8 participants that vary with respect to gender, country, clinical background, and number of systematic reviews they have ever led. To achieve saturation, we will continue to enroll participants until they do not identify any new issues or problems about instrument.

We will send them the instrument and the manual one week before the interview and ask them to read these documents before the interview.

During the interviews, we will collect participants' concerns or suggestions on the instrument and the manual. We will record and transcribe the interviews. We will extract participants' concerns or suggestions and any problems they may have if using this instrument.

As soon as we find a clear problem in these documents, we will make a correction immediately and continue the use testing with the corrected version.

Table 1. Possible categories of risk of bias methodological papers*

Classification	Subclassification
Overall risk of bias	Studies addressing categorization/dimension of bias in RCTs Studies illustrating approach to incorporating risk of bias assessment of RCTs into meta-analysis Studies about RCT risk of bias instrument
Random sequence generation	Empirical study illustrating how random sequence generation influences results Theoretical discussion about why random sequence generation is important Studies introducing/criticizing specific randomization approach Studies about baseline imbalance
Allocation concealment	Empirical study illustrating how allocation concealment influences results Theoretical discussion about why allocation concealment is important Studies introducing/criticizing specific allocation concealment approach Studies about deciphering allocation sequences Studies about reporting of allocation concealment
Blinding	Empirical study illustrating how blinding influences results Theoretical discussion about why blinding is important Introducing/criticizing blinding method Interpretation of blinding terminology/ reporting of blinding status Testing the successful of blinding

Missing outcome data	Empirical study illustrating how missing data influences results Methodological paper introducing how to deal with missing outcome data in RCTs Methodological paper introducing how to deal with missing outcome data in meta-analyses Systematic survey investigating how RCTs dealing with missing outcome data Systematic survey investigating how meta-analyses dealing with missing outcome data Strategies for identifying missing data Relationship between missing data and intention to treat
Intention to treat	Methodological papers advocating intention to treat analysis Meaning/definition of intention to treat Empirical study illustrating how deviation from intention to treat analysis influences results Effect of adhering to intervention Post-randomization exclusion
Selective reporting	Empirical study illustrating how selective reporting influences results Assessing selective reporting in RCTs General discussion about selective reporting Categorization of selective reporting in RCTs
Stop early for benefit	Empirical study illustrating how stop early for benefit influences results Theoretical discussion about why stop early for benefit is or is not a risk of bias issue Simulation study illustrating how stop early for benefit influences results Empirical study examining the prevalence of stop early for benefit or reviews how to deal with stop early for benefit
Center status	Empirical study illustrating how center status influences results
Funding	Empirical study illustrating how funding influences results

	Theoretical discussion about why funding is or is not a risk of bias issue
Empirical study/ meta-epidemiological study involving more than one risk of bias issue	Empirical study illustrating how items influence results Empirical study examining the prevalence of several items
Making judgement when poor reporting	Identify which risk of bias item(s) is/are examined
Cluster/crossover RCTs	NA

*If authors stated explicitly or implicitly indicated that what they addressed is a risk of bias issue, the papers are eligible.

References

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Appendix 2. Details of the user testing of the ROBUST-RCT

Table 1. Characteristics of the junior systematic reviewers in the user-testing

Characteristic	**Number of participants (total 15)**
Female	7
Country	
Canada	7
China	3
India	2
Switzerland	2
US	1
Clinical background	
Physician	7
Pharmacist	2
Dietitian	1
No clinical background	5
Student status	
PhD student	2
Master student	6
Undergraduate student	3
Not student	4
Number of systematic reviews in which they have assessed risk of bias of randomized trials	
1-2	10
3-5	3
>5	2

Table 2. Trials that used in user testing with junior systematic reviewers

1. Chochinov HM, Kristjanson LJ, Breitbart W, et al. Effect of dignity therapy on distress and end-of-life experience in terminally ill patients: a randomised controlled trial. Lancet Oncol. 2011 Aug;12(8):753-62. doi: 10.1016/S1470-2045(11)70153-X.
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Table 3. Summary of feedback from user testing with junior systematic reviewers and resulting changes

Items	Feedback or problem found in the user testing	Resulting changes
Instrument version A (Ultimate first step)		
Item 1: Was the allocation sequence adequately generated	1. One out of the first five reviewers got confused by the example "shuffling cards or envelopes" in definitely yes.	1. Since we provide a number of examples, and shuffling cards or envelopes is seldom used, the panel removed this specific example.
	2. The initial instruction for probably yes was "Trial was described as 'randomized' without further details regarding the method of generating the allocation sequence, and the text described the allocation concealment method". As long as the trial reported using central allocation, drug containers or envelopes, the criterion is satisfied (e.g., no need to explicitly state sequentially numbered opaque sealed envelopes). One out of the second five reviewers had problem assessing this correctly.	2. We added a parenthesis as follows: "Trial was described as 'randomized' without further details regarding the method of generating the allocation sequence, and the text described the allocation concealment method (stated using central allocation, drug containers, or envelopes)".
Item 2: Was the allocation adequately concealed	1. Initially we had independent central allocation and independent pharmacy-controlled randomization in definitely yes; and central allocation or pharmacy-controlled randomization without specification of independent in probably yes ("independent" means that the investigator generating the sequence is different from the investigator enrolling and assigning participants). Four out of the first five reviewers had problem understanding "independent". This led to mistakes in assessment (in the two trials where this was an issue, they rated as definitely yes when it should have been probably yes and rated as probably yes when it should have been definitely yes).	1. The panel decided to omit the independent issue. Thus, in the latest instrument, central allocation and pharmacy-controlled randomization should be assessed as definitely yes.
	2. For the envelopes approach, two out of the second five reviewers	2. We made the difference in italics.

	assessed wrongly because they ignored the difference between "sequentially numbered opaque sealed envelopes with evidence that they were opened sequentially" in definitely yes and "sequentially numbered, opaque, sealed envelopes without further details" in probably yes.	
	3. Two out of the second five reviewers thought the difference between probably yes and probably no was not very clear (probably yes: “Drug trial in which participants and healthcare providers were blinded...”; probably no: “Unblinded drug trial or non-drug trial...")	3. We made the difference bold.
Item 3: Were participants blinded	No problem found.	No change.
Item 4: Were healthcare providers blinded	No problem found.	No change.
Item 5: Were outcome assessors blinded	Two out the first five reviewers assessed wrongly because they didn’t understand that, when the participant completed a questionnaire, participants were the outcome assessors.	We added “When the outcome is participants self-report, e.g., completing a questionnaire (such as quality of life, disability index), participants are the outcome assessors”.
Item 6: Section 1. Extract the number of participants who were not included in analysis in each group Section 2. Was the overall percentage of participants not included in analysis acceptably low	1. In the instrument used in the first five interviews, the initial item 6 was: Section 1. Extract number of participants with missing outcome data in each group Section 2. Did the trial achieve an acceptably low percentage of participants with missing outcome data The initial item 7 was: For participants whose outcome data were available, were all or almost all of them analyzed in the group to which they were randomized	1. The panel split the initial item 7 into two issues: per-protocol analysis and as-treated analysis. The panel combined the per-protocol analysis issue with the initial item 6 together, as the current item 6. Thus, the current item 6 addresses the participants who were not included in analysis irrespective of the reasons. The following user testing suggested the new item works well. The panel added the as-treated analysis issue as an optional item (optional item 6) since it happens rarely.

	Four out of the first five reviewers, when they assessing different trials, assessed wrongly for the initial item 7 - they had difficulty differentiating the participants who were left out of analysis because of decisions by the trialists to not include non-compliant individuals (i.e., should be addressed in initial item 7), and the participants cannot be included in analysis because their outcome data were not available to the trialists (should be addressed in initial item 6).	
	2. Reviewers had difficulty getting the number of participants not included in analysis correctly for time-to-event outcomes.	2. We added "For time-to-event outcomes, also count the participants who were censored because of missing follow-up data in 'N not analyzed'."
Overall experience in applying version A (ultimate first step)	All reviewers thought items and instructions in version A are clear and easy to apply.	No change.
Instrument version B (Ultimate second step)		
Item 1: Judge risk of bias related to sequence generation	Same as version A.	Same as version A.
Item 2: Judge risk of bias related to allocation concealment	Same as version A.	Same as version A.
Item 3: Judge risk of bias related to blinding of participants	Reviewer suggested bolding the adverbs of degree: very unlikely, unlikely, likely and very likely.	We made the words very unlikely, unlikely, likely and very likely bold.
Item 4: Judge risk of bias related to blinding of healthcare	No problem found.	No change.

providers		
Item 5: Judge risk of bias related to blinding of outcome assessors	No problem found.	No change.
Item 6: Section 1. Extract the number of participants who were not included in analysis in each group Section 2. Judge risk of bias related to the overall percentage of participants not included in analysis	Same as version A.	Same as version A.
Overall experience in applying version B (ultimate second step)	All reviewers thought items and instructions in version B are clear. Four reviewers thought version B makes more sense because they have get used to assess risk of bias directly.	No change.

Table 4. Characteristics of the systematic review experts in the user-testing

Characteristic	Number of participants (total 8)
Female	5
Country	
Canada	2
China	2
Germany	2
Australia	1
UK	1
Clinical background	
Physician	4
Nurse	1
No clinical background	3

Table 5. Summary of feedback from user testing with systematic review experts and resulting changes

Document	Feedback or problem found in the user testing	Resulting changes
Instrument	Participants questioned the rationale for using only version A. One person suggested to make the two instrument versions as two steps for assessing risk of bias: first step is evaluating what happened, and second step is judging risk of bias based on what happened.	We changed to adopt the two-step approach: for each core item, there are two steps: first step is evaluating what happened, and second step is judging risk of bias based on what happened.
	One participant suggested to add in each item a space for “support for judgement”.	We added a space for “support for judgement” for each item.
	One participant suggested to add a space for systematic reviewers to add optional items in the instrument.	We added a space for optional items in the last page of the instrument.
	Tow participants suggested to think of a way to let systematic reviewers enter their risk of bias assessment results and generate a risk of bias assessment table for all trials in the systematic review which can be put in the systematic review manuscripts.	We developed an excel where systematic reviewers can enter their risk of bias assessment results and thus generate a risk of bias assessment table for all trials in the systematic review (appendix 5).
Manual	One participant suggested to rewrite the titles for optional items to make them easier to understand. For instance, optional item 1 "Baseline prognostic factors balance vs. imbalance", change to "Whether the baseline prognostic factors balanced between groups ".	We rewrote the titles for optional items.
	One participant suggested to add some examples for behavioral or psychological trials in the manual.	We added examples for behavioral and psychological trials.
	One participant suggested to change the sentence “Trials usually do not report number of participants who were not included in analysis for individual outcomes” to “Trials may not report number of participants who were not included in analysis for individual outcomes”.	We changed to sentence to “Trials may not report number of participants who were not included in analysis for individual outcomes”.

Appendix 3. Panel's initial decisions on item selection and format of the instrument

Table 1. Judgment regarding the extent to which items in category 1 (items that the majority judged as addressing risk of bias in a survey) meet the six item selection criteria and item selection decisions

Items	Item selection criteria						Item selection decision
	Clearly risk of bias rather than others	Theoretical or logical argument for why the item is important	Information required to make judgement is commonly reported in trials	Non-expert reviewers can make the judgment easily	Problem occurs more often than rarely	Empirical evidence supports item influence on effect estimates	
Random sequence generation	√ (category 1)	√	√	√	√	Overestimation (Moderate certainty)	**Include as core item 1**
Allocation concealment	√ (category 1)	√	√	√	√	Overestimation (Moderate certainty)	**Include as core item 2**
Blinding of participants	√ (category 1)	√	√	√	√	Any outcomes: Very uncertain Patient-reported outcomes: Overestimation (Moderate certainty) Observer-reported or objective outcomes: Very uncertain	**Include as core item 3**
Blinding of healthcare providers	√ (category 1)	√	√	√	√	Very uncertain	**Include as core item 4**
Blinding of data collectors	√ (category 1)	√	×	×	√	Very uncertain	**Exclude**
Blinding of outcome assessors	√ (category 1)	√	√	√	√	Any outcomes: Very uncertain Objective outcomes: Very uncertain Subjective outcomes: Overestimation (High certainty)	**Include as core item 5**
Blinding of data analysts	√ (category 1)	√	×	×	√	Very uncertain	**Exclude**
Whether the outcome assessment or data collection differed between groups	√ (category 1)	√	×	×	×	No evidence	**Include as optional item 3**
Whether the follow-up time, frequency, or intensity of outcome assessment differed between groups	√ (category 1)	√	×	×	×	No evidence	**Include as optional item 4**
Missing outcome data	√ (category 1)	√	√	√	√	Underestimation (Low certainty)	**Initially include as a core item. Afterward, based on use**

			testing feedback, the panel combined the missing outcome data issue and per-protocol analysis issue together as a new core item (core item 6).

Table 2. Decisions for items in category 2 (items that the majority judged as not addressing risk of bias in a survey)*

Items	Item selection decision	Reasons for excluding
Whether the outcome measurement was reliable (i.e., reliability of outcome measurement)	**Exclude**	Imprecision (random error) rather than risk of bias.
Whether the follow-up time was adequate to identify the outcome of interest	**Exclude**	Indirectness (applicability) rather than risk of bias.
Whether the sample size was big enough	**Exclude**	Imprecision (random error) rather than risk of bias.
Whether the sampling approach (approach to selecting participants from whole population) was appropriate	**Exclude**	Indirectness (applicability) rather than risk of bias.
Whether there was conflict of interest	**Exclude**	Not directly related to risk of bias – may influence effect estimate but must through other mechanisms.
Whether there was funding	**Exclude**	Not directly related to risk of bias – may influence effect estimate but must through other mechanisms.
Whether the results were comparable for all trial sites	**Exclude**	Not related to risk of bias.
Whether there was run-in period before randomization	**Exclude**	Not related to risk of bias.
Whether the inclusion and exclusion criteria were appropriate	**Exclude**	Indirectness (applicability) rather than risk of bias.

*All items in category 2 were excluded.

Table 3. Judgment regarding the extent to which items in category 3 (items with substantial disagreement regarding addressing risk of bias or not) meet the six item selection criteria and item selection decisions

Items	Item selection criteria						Item selection decision
	Clearly risk of bias rather than others	Theoretical or logical argument for why the item is important	Information required to make judgement is commonly reported in trials	Non-expert reviewers can make the judgment easily	Problem occurs more often than rarely	Empirical evidence supports item influence on effect estimates	
Whether the baseline prognostic factors were balanced between groups	×	√	√	×	?	Very uncertain	**Include as optional item 1**
Whether the co-interventions were balanced between groups in blinded trials	×	√	×	×	?	Overestimation (Low certainty)	**Include as optional item 2**
Whether the interventions were implemented as intended	×	√	×	×	√	Very uncertain	**Exclude**
Whether the participants were adhered to the assigned interventions (nonadherence could be imperfect compliance, cessation, crossover, or switch to another active intervention)	×	×	×	×	√	Very uncertain	**Exclude**
Whether the outcome measurement method was valid (i.e., validity of outcome measurement)	×	√	×	×	×	No evidence	**Include as optional item 5**
Whether there was non-differential outcome measurement error (errors that were unrelated to intervention assignment)	×	√	×	×	×	No evidence	**Exclude**
Whether the method of outcome measurement or data collection was	×	×	×	×	×	No evidence	**Exclude**

sensitive to plausible intervention effects							
Whether there was selective reporting	×	√	×	×	×	Very uncertain	**Include as optional item 7**
Intention-to-treat analysis	×	√	×	√	√	Very uncertain	**Initially include as a core item. Afterward, based on use testing feedback, the panel split this item into two types of deviation: per-protocol analysis and as-treated analysis. The panel combined the per-protocol analysis issue with the missing outcome data issue together as a new core item (core item 6).** **The panel added the as-treated analysis issue as optional item 6: "when investigators conducted an as-treated analysis, was the percentage of participants not analyzed in the groups to which they were randomized sufficiently low".**
Whether the trial was terminated early	×	√	√	√	√	Stop early for benefit: Overestimation (Moderate certainty)	**Include as optional item 8.** **The panel rewrote the optional item as "whether the trial was terminated early for benefit". Although trials that terminated early for futility may underestimate effect, the underestimation was much less dramatic. As for trials terminated early for harm, although they would overestimate the harm, if a trial showed an early signal of harm with the intervention it may be unethical to enroll more patients into the intervention group.**

Table 4. Initial instrument version A (finally changed to the first step for assessing risk of bias)

Task of systematic reviewers who assess individual trials (i.e., front-line systematic reviewers)	Evaluate what happened for each item Item 1: Was the allocation sequence adequately generated Item 2: Was the allocation adequately concealed Item 3: Were participants blinded Item 4: Were healthcare providers blinded Item 5: Were outcome assessors blinded Item 6 - Section 1 (necessary): Extract the number of participants who were not included in analysis in each group - Section 2 (optional): Was the overall percentage of participants not included in analysis acceptably low
Response options	Definitely Yes; Probably Yes; Probably No; Definitely No (except for item 6 section 1)
Who make risk of bias judgement	Not front-line systematic reviewers; rather, systematic review team members who assess GRADE certainty of evidence across trials (usually review leaders)

Table 5. Initial instrument version B (finally changed to the second step for assessing risk of bias)

Task of systematic reviewers who assess individual trials (i.e., front-line systematic reviewers)	Judge risk of bias related to each item Item 1: Judge risk of bias related to sequence generation Item 2: Judge risk of bias related to allocation concealment Item 3: Judge risk of bias related to blinding of participants Item 4: Judge risk of bias related to blinding of healthcare providers Item 5: Judge risk of bias related to blinding of outcome assessors Item 6 - Section 1 (necessary): Extract the number of participants who were not included in analysis in each group - Section 2 (optional): Judge risk of bias related to the overall percentage of participants not included in analysis
Response options	Definitely Low; Probably Low; Probably High; Definitely High (except for item 6 section 1)
Who make risk of bias judgement	Front-line systematic reviewers

Risk Of Bias instrument for Use in SysTematic reviews-for Randomized Controlled Trials (ROBUST-RCT)

Study reference:

State the outcome(s) that are being assessed for risk of bias:

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Item 1: Random sequence generation

Step 1: **Was the allocation sequence adequately generated***	**Step 2:** **Judge risk of bias related to sequence generation***	**Instructions**
☐ **Definitely Yes**	☐ **Definitely Low**	Trial explicitly stated use of an adequate method of generating the random allocation sequence. Examples include: random number table; random number generator; throwing dice; drawing of lots; minimization.
☐ **Probably Yes**	☐ **Probably Low**	Trial was described as "randomized" without further details regarding the method of generating the allocation sequence, and: - the text mentioned simple randomization, block randomization, or stratified randomization; or - the text described the allocation concealment method (stated using central allocation, drug containers, or envelopes).
☐ **Probably No**	☐ **Probably High**	Trial was described as "randomized" without further details regarding the method of generating the allocation sequence, and it does not meet any of the criteria for "Probably Yes/Low".
☐ **Definitely No**	☐ **Definitely High**	Trial used a non-randomized allocation sequence that would be recognized as "quasi-randomization". Examples include: allocation based on dates of birth or admission; patient's hospital record number; alteration or rotation; allocation decided by clinicians or participants; allocation based on the results of a laboratory test; allocation by availability of the intervention.

*Instructions for step 1 and step 2 are the same because no additional judgement is involved.

Support for judgement:

Item 2: Allocation concealment

Step 1: **Was the allocation adequately concealed***	**Step 2:** **Judge risk of bias related to allocation concealment***	**Instructions**
☐ **Definitely Yes**	☐ **Definitely Low**	Trial used a clearly satisfactory allocation concealment method: central allocation (e.g., telephone, web-based); pharmacy-controlled randomization (stated using sequentially numbered sealed drug containers); trial explicitly stated using sequentially numbered opaque sealed envelopes *with evidence that they were opened sequentially*.
☐ **Probably Yes**	☐ **Probably Low**	• Trial explicitly stated using sequentially numbered, opaque, sealed envelopes *without further details*. • *Drug trial in which participants and healthcare providers were blinded*: - with no further information on the allocation concealment method; or - stated using envelopes but it remains unclear whether the envelopes were sequentially numbered opaque and sealed; or - stated using drug containers but it remains unclear whether the drug containers were sequentially numbered and sealed.
☐ **Probably No**	☐ **Probably High**	• *Unblinded drug trial or non-drug trial*: - with no further information on the allocation concealment method; or - stated using envelopes but it remains unclear whether the envelopes were sequentially numbered opaque and sealed; or - stated using drug containers but it remains unclear whether the drug containers were sequentially numbered and sealed.
☐ **Definitely No**	☐ **Definitely High**	• Trial used an open random allocation schedule. • Trial used a non-randomized allocation sequence that would be recognized as “quasi-randomization”.

*Instructions for step 1 and step 2 are the same because no additional judgement is involved.

Support for judgement:

Item 3: Blinding of participants

Step 1: Were participants blinded

☐ **Definitely Yes**	Trial explicitly stated that participants were blinded.
☐ **Probably Yes**	No explicit statement about blinding of participants, and: - it is a placebo-controlled drug trial; or - it is an active control drug trial (A vs. B) and mention of "double dummy" or that medications were identical or matched; or - trial was described as "single blinded", "double blinded" or "triple blinded", and the best judgment is that one of the blinded groups is the participants; or - participants not capable of distinguishing if they are receiving active or control intervention (e.g., neonates, severely demented).
☐ **Probably No**	No explicit statement about blinding of participants, and: - it is an active control drug trial (A vs. B) and no mention of "double dummy" or that medications were identical or matched; or - it is a non-drug trial; or - trial was described as "single blinded" and the best judgement is that the single blinded group is someone other than the participants.
☐ **Definitely No**	• Trial explicitly stated that participants were not blinded. • Trial was described as "open-label" or "unblinded".

Support for judgement:

Step 2: Judge risk of bias related to blinding of participants

Issues to consider:

i) Were participants blinded (step 1)

ii) If unblinded, how likely unblinding of participants has influenced the outcome

- *How likely are participants expectations regarding effect of intervention to have influenced the outcome*
- *How likely are participant-initiated co-interventions to have influenced the outcome*

☐ **Definitely Low**	• Participants were definitely blinded; OR • Unblinding of participants *very unlikely* to have influenced the outcome because: very unlikely participants expectations regarding effect of intervention have influenced the outcome and very unlikely participant-initiated co-interventions have influenced the outcome.
☐ **Probably Low**	• Participants were probably blinded; OR • Unblinding of participants *unlikely* to have influenced the outcome because: unlikely participants expectations regarding effect of intervention have influenced the outcome and unlikely participant-initiated co-interventions have influenced the outcome.
☐ **Probably High**	Participants were definitely or probably not blinded, AND unblinding of participants *likely* to have influenced the outcome because participants expectations regarding effect of intervention likely to have influenced the outcome or participant-initiated co-interventions likely to have influenced the outcome.
☐ **Definitely High**	Participants were definitely or probably not blinded, AND unblinding of participants *very likely* to have influenced the outcome through participants expectations regarding effect of intervention or through participant-initiated co-interventions.

Support for judgement:

Item 4: Blinding of healthcare providers

Step 1: Were healthcare providers blinded

☐ **Definitely Yes**	Trial explicitly stated that healthcare providers were blinded.
☐ **Probably Yes**	No explicit statement about blinding of healthcare providers, and: - it is a placebo-controlled drug trial; or - it is an active control drug trial (A vs. B) and mention of "double dummy" or that medications were identical or matched; or - trial was described as "single blinded", "double blinded" or "triple blinded", and the best judgment is that one of the blinded groups is the healthcare providers.
☐ **Probably No**	No explicit statement about blinding of healthcare providers, and: - it is an active control drug trial (A vs. B) but no mention of "double dummy" or that medications were identical or matched; or - it is a non-drug trial; or - trial was described as "single blinded" and the best judgement is that the single blinded group is someone other than the healthcare providers.
☐ **Definitely No**	• Trial explicitly stated that healthcare providers were not blinded. • Trial was described as "open-label" or "unblinded".

Support for judgement:

Step 2: Judge risk of bias related to blinding of healthcare providers

Issues to consider:

i) Were healthcare providers blinded (step 1)

ii) *If unblinded, how likely unblinding of healthcare providers has influenced the outcome*

- *How likely healthcare provider-initiated co-interventions have influenced the outcome*

☐ **Definitely Low**	Healthcare providers were definitely blinded.
☐ **Probably Low**	• Healthcare providers were probably blinded; OR • Unblinding of healthcare providers unlikely to have influenced the outcome because: - Unlikely there is any healthcare provider-initiated co-intervention that could potentially influence the outcome; or - Investigators have documented all healthcare provider-initiated co-interventions that could potentially influence the outcome and demonstrated similarity in use of all these co-interventions between groups.
☐ **Probably High**	• Healthcare providers were definitely or probably not blinded, AND unblinding of healthcare providers likely to have influenced the outcome because there are healthcare provider-initiated co-interventions that could potentially influence the outcome.
☐ **Definitely High**	• Healthcare providers were definitely or probably not blinded, AND unblinding of healthcare providers very likely to have influenced the outcome because there are healthcare provider-initiated co-interventions that could potentially influence the outcome and investigators have documented dissimilarity in any of these co-interventions between groups.

Support for judgement:

Item 5: Blinding of outcome assessors

Step 1: Were outcome assessors blinded

When the outcome is participants self-report, e.g., completing a questionnaire (such as quality of life, disability index), participants are the outcome assessors.

☐ **Definitely Yes**	Trial explicitly stated that the outcome assessors or adjudicators (people making the measurement or assessment) were blinded.
☐ **Probably Yes**	No explicit statement about blinding of outcome assessors or adjudicators, and: - it is a placebo-controlled drug trial; or - it is an active control drug trial (A vs. B) and mention of "double dummy" or that medications were identical or matched; or - trial was described as "single blinded", "double blinded" or "triple blinded", and the best judgement is that one of the blinded groups is the outcome assessors.
☐ **Probably No**	No explicit statement about blinding of outcome assessors or adjudicators, and: - it is an active control drug trial (A vs. B) but no mention of "double dummy" or that medications were identical or matched; or - it is a non-drug trial; or - trial was described as "single blinded" and the best judgement is that the single blinded group is someone other than the outcome assessors.
☐ **Definitely No**	• Trial explicitly stated that the outcome assessors or adjudicators were not blinded. • Trial was described as "open-label" or "unblinded".

Support for judgement:

Step 2: Judge risk of bias related to blinding of outcome assessors

Issues to consider:

i) Were outcome assessors blinded (step 1)

ii) If unblinded, how likely unblinding of outcome assessors has influenced the outcome assessment

- *Degree of judgement/subjectivity involved in the outcome assessment (more judgment/subjectivity more likelihood of bias)*

☐ **Definitely Low**	• Outcome assessors were definitely blinded; OR • Outcome is all-cause mortality.
☐ **Probably Low**	• Outcome assessors were probably blinded; OR • Unblinding of outcome assessors likely could not have influenced the outcome assessment because the outcome assessment involves minimal judgement (other objective outcomes, e.g., laboratory measurement, hospital admission, mechanical ventilation).
☐ **Probably High**	Outcome assessors were definitely or probably not blinded, AND unblinding of outcome assessors likely could have influenced the outcome assessment because the outcome assessment involves some judgment (e.g., cause-specific mortality).
☐ **Definitely High**	Outcome assessors were definitely or probably not blinded, AND unblinding of outcome assessors could have influenced the outcome assessment because the outcome assessment involves considerable judgment by participant or adjudicator (e.g., symptoms and symptom scores, quality of life, seizure occurrence).

Support for judgement:

Item 6: Outcome data not included in analysis

Step 1: Extract the number of participants who were not included in analysis in each group

Note: For time-to-event outcomes, also count the participants who were censored because of missing follow-up data in 'N not analyzed'.

	N not analyzed	N total (usually N randomized)	Percentage not analyzed (N not analyzed / N total)
Intervention group _____			
Control group _____			
Overall			

Step 2: Judge risk of bias related to the overall percentage of participants not included in analysis

Issue to consider: Was the overall percentage of participants not included in analysis acceptably low

Note: Systematic review teams need to set and fill in the threshold for each response option. See manual for instructions and example thresholds.

☐ **Definitely Low**	Percentage of participants not included in analysis is < %
☐ **Probably Low**	• Percentage of participants not included in analysis is % to < % • If the trial did not mention whether there were participants not included in analysis, a substantial loss to follow-up is unlikely (e.g., ICU mortality)
☐ **Probably High**	• Percentage of participants not included in analysis is % to < % • If the trial did not mention whether there were participants not included in analysis, a substantial loss to follow-up is likely (e.g., 1-year quality of life)
☐ **Definitely High**	Percentage of participants not included in analysis is ≥ %

Support for judgement:

Which if any optional item(s) to assess

We provide eight optional items (see manual).

We strongly recommend that the systematic review teams consider each of the optional items carefully and judge whether they are relevant to their particular reviews.

If systematic review teams decide to include optional item(s), specify the optional items that need to be considered and provide instructions for assessing the items (see examples in manual).

Optional item	Instruction	Judgement	Support for judgement

Study	Outcome(s)	Item 1: Random sequence gen
		Step 1: Was the allocation sequence adequately generated

eration		Item 2: Allocation concealmen
Step 2: Judge risk of bias related to sequence generation	Support for judgement	**Step 1: Was the allocation adequately concealed**

t		Item 3: Blinding of participants
Step 2: Judge risk of bias related to allocation concealment	Support for judgement	**Step 1: Were participants blinded**

s		
Support for judgement	**Step 2: Judge risk of bias related to blinding of participants**	Support for judgement

Item 4: Blinding of healthcare providers		
Step 1: Were healthcare providers blinded	Support for judgement	**Step 2: Judge risk of bias related to blinding of healthcare providers**

	Item 5: Blinding of outcome assessors	
Support for judgement	**Step 1: Were outcome assessors blinded**	Support for judgement

		Item 6: Outcome data not included in anal	
Step 2: Judge risk of bias related to blinding of outcome assessors	Support for judgement	**Step 1: Extract the number of participants**	
		Intervention group	
		N not analyzed	N total

ysis				
who were not included in analysis in each group				
	Control group			Overall
% not analyzed	N not analyzed	N total	% not analyzed	N not analyzed

		Step 2: Judge risk of bias related to the overall percentage of participants not included in analysis
N total	% not analyzed	

Support for judgement

*Add optional item(s) if there is any

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Manual

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Two steps for assessing risk of bias

ROBUST-RCT suggests two steps for assessing risk of bias for each core item.

The first step is evaluating what happened, that is, whether the methodological safeguard has been implemented (e.g., step 1 of item 3 is judging if participants were blinded). For all but the last item, response options include definitely yes, probably yes, probably no, and definitely no.

The second step is judging risk of bias based on what happened (e.g., step 2 of item 3 is judging risk of bias related to blinding of participants). It requires systematic review team members to decide the extent to which any deficits in instituting methodological safeguards actually resulted in risk of bias. Response options include definitely low, probably low, probably high, and definitely high risk of bias. Table 1 presents the two steps for each core item.

Systematic reviewers assessing individual trials (i.e., risk of bias assessors) can complete both steps. However, for core items 1-5, if the risk of bias assessors that the systematic review team recruits are less experienced and may face difficulty in judging risk of bias, review leaders may ask the risk of bias assessors to complete only step 1 and leave step 2 to the reviewers with more experience.

For core item 6 'outcome data not included in analysis', there are two approaches to addressing risk of bias. One is deciding the risk of bias associated with this item for each individual trial. In this case, systematic review teams will need to set the missing percentage threshold for each response option for the step 2 of item 6 (see instructions for setting thresholds in page 23). Risk of bias assessors will determine the percentage of people not included in analysis and where that percentage falls in the risk of bias categories.

Another approach for core item 6 involves systematic review teams assessing risk of bias associated with missing data across the entire body of evidence at the meta-analysis level.[1 2] The process of doing so begins with a complete-case analysis followed by an analysis imputing data for participants in each trial who were not included in the analysis (see details in page 23).[1 2] In this case, risk of bias assessors can complete only step 1 in which they extract the number of participants who were not included in analysis.

Table 1. ROBUST-RCT core items and two steps

	Step 1 Evaluate what happened	Step 2 Judge risk of bias
Item 1 Random sequence generation	Was the allocation sequence adequately generated	Judge risk of bias related to sequence generation
Item 2 Allocation concealment	Was the allocation adequately concealed	Judge risk of bias related to allocation concealment
Item 3 Blinding of participants	Were participants blinded	Judge risk of bias related to blinding of participants
Item 4 Blinding of healthcare providers	Were healthcare providers blinded	Judge risk of bias related to blinding of healthcare providers
Item 5 Blinding of outcome assessors	Were outcome assessors blinded	Judge risk of bias related to blinding of outcome assessors
Item 6 Outcome data not included in analysis	Extract the number of participants who were not included in analysis in each group	Judge risk of bias related to the overall percentage of participants not included in analysis
Response options	Definitely Yes; Probably Yes; Probably No; Definitely No (except for item 6)	Definitely Low; Probably Low; Probably High; Definitely High

State the outcome(s) that are being assessed for risk of bias

Systematic review teams need to specify the outcome(s) that are being assessed for risk of bias (fill in the first page of the instrument).

Study reference:

State the outcome(s) that are being assessed for risk of bias:

Systematic review teams need to consider whether to direct systematic reviewers to make one assessment for all outcomes, a group of outcomes, or for each outcome separately (Table 2).

Table 2. Considerations regarding assessing all outcomes, a group of outcomes, or each outcome separately

Items	Considerations
Item 1 Random sequence generation Item 2 Allocation concealment	Same for all outcomes; can assess once for all outcomes.
Item 3 Blinding of participants Item 4 Blinding of healthcare providers	For step 1 judging if participants/healthcare providers were blinded, judgement is likely to be the same for all outcomes; if the systematic reviewers assessing individual trials need to complete only step 1, they could assess once for all outcomes. If the systematic reviewers assessing individual trials need to complete both steps, the review team should consider for which outcomes extent of risk of bias is likely to differ. When this is likely, the safest approach is to ask systematic reviewers to rate each outcome separately.
Item 5 Blinding of outcome assessors	If the systematic reviewers assessing individual trials need to complete only step 1, the review team should consider whether the outcome assessors were the same for different outcomes. If not, they should consider whether it is plausible that blinding may differ across the different outcome assessors. If that is likely, the review team may choose to assess each outcome separately. If the systematic reviewers assessing individual trials need to complete both steps, the review team should consider for which outcomes extent of risk of bias is likely to differ. When this is likely, the safest approach is to ask systematic reviewers to rate each outcome separately.
Item 6 Outcome data not included in analysis	Trials may not report number of participants who were not included in analysis for individual outcomes. In such instances, rating for item 6 will likely be identical for different outcomes. The review team should, however, consider the possibility that trialists may have reported participants not included in analysis separately for each outcome, in which case item 6 should be rated separately. The

	review leaders may want to check this themselves and then specify for risk of bias assessors which outcomes should be assessed separately, or direct risk of bias assessors to be themselves alert to these issues. They may want to consider the possibility that although loss to follow-up is specified for only a single outcome (e.g., mortality), loss to follow-up although unspecified may be greater for another outcome (e.g., quality of life).

Item 1 Random sequence generation

Step 1: Was the allocation sequence adequately generated
Step 2: Judge risk of bias related to sequence generation
Instructions for step 1 and step 2 are the same.

Definitely Yes/Low
Trial explicitly stated use of an adequate method of generating the random allocation sequence. Examples include: random number table; random number generator; throwing dice; drawing of lots; minimization.
Explanation:
Adequate method of generating the random allocation sequence refers to the method that incorporates a random element and thus can generate a random and unpredictable sequence.
Minimization is a method of ensuring intervention groups are closely similar for multiple prognostic factors, even in small trials.[3] Using minimization, the first participant is allocated randomly. For each subsequent participant, investigators determine assignment to the intervention that would lead to better balance between the groups over all the identified prognostic factors.
Example: A trial stated "Randomization was performed with a computer-generated allocation sequence…".[4]

Probably Yes/Low
Trial was described as "randomized" without further details regarding the method of generating the allocation sequence, and:
- **the text mentioned simple randomization, block randomization, or stratified randomization; or**
- **the text described the allocation concealment method (stated using central allocation, drug containers, or envelopes).**

Explanation:
Sometimes, in trial reports, authors described the trials as "randomized" or stated "randomly allocated". They did not, however, explicitly report the method they used for generating the allocation sequence (i.e., did not specify any of the methods in 'Definitely Yes/Low' or 'Definitely No/High').

Simple randomization (unrestricted randomization) means allocating each participant at random independently with no constraints.[5] If a trial did not specify the method trialists used for generating the allocation sequence, but it reported that it adopted simple or unrestricted randomization approach, then trialists probably have used an adequate method of generating the random sequence.

Block randomization (one type of restricted randomization) is used when trialists want to ensure that the numbers of participants in intervention and control groups reach a particular ratio (e.g., 1:1).[5] If a trial reported that it adopted block randomization or mentioned blocking, although it did not specify the method used for generating the allocation sequence for each block, it is likely that trialists used an adequate method of generating the random sequence.

Stratified randomization is used when trialists want to ensure that the participants in intervention and control groups are balanced regarding a particular prognostic factor.[5] If a trial did not specify the method trialists used for generating the allocation sequence, but it reported that it adopted stratified randomization, then trialists probably have used an adequate method of generating the random sequence.

Allocation concealment method: If authors described the trial as "randomized" or stated "randomly allocated", and reported using central allocation, pharmacy-controlled randomization (or using drug containers), or envelopes, trialists probably have generated the sequence adequately.
Example:

1. A trial stated "Patients were randomly allocated into two groups using blocked randomization" but it did not report how they generated the random sequence.[6]
2. A trial stated "Randomization was by sealed envelope using the permuted block design" but it did not report how they generated the random sequence.[7]

Probably No/High

Trial was described as "randomized" without further details regarding the method of generating the allocation sequence, and it does not meet any of the criteria for "Probably Yes/Low".
Example: A trial stated "patients were randomly allocated" without further details regarding the sequence generation method, and it did not mention simple, block, or stratified randomization, and it did not describe the allocation concealment method.[8]

Definitely No/High

Trial used a non-randomized allocation sequence that would be recognized as "quasi-randomization". Examples include: allocation based on dates of birth or admission; patient's hospital record number; alteration or rotation; allocation decided by clinicians or participants; allocation based on the results of a laboratory test; allocation by availability of the intervention.
Explanation: *"Quasi-randomization"* methods are actually non-randomized allocation methods because they are not based on a random/chance process and they cannot generate an unpredictable allocation sequence.[9]
Example: A trial stated "Patients were randomized to each respective study arm by date of admission. Treatment regimens alternated every other month. Patients admitted into the unit during odd-numbered months received famotidine 20 mg IV every 12 h; patients admitted during even numbered months received lansoprazole suspension 30 mg suspended in 10 ml of an 8.4% sodium bicarbonate solution or apple juice, and administered via NG tube daily."[10]

Item 2 Allocation concealment

Step 1: Was the allocation adequately concealed
Step 2: Judge risk of bias related to allocation concealment
Instructions for step 1 and step 2 are the same.

Definitely Yes/Low

Trial used a clearly satisfactory allocation concealment method: central allocation (e.g., telephone, web-based); pharmacy-controlled randomization (stated using sequentially numbered sealed drug containers); trial explicitly stated using sequentially numbered opaque sealed envelopes with evidence that they were opened sequentially.

Explanation:

Allocation concealment means trialists take procedures to prevent participants and investigators involved in the enrollment and assignment from foreknowing or predicting the forthcoming allocations.

Central allocation (or remote allocation/randomization) means the allocation is controlled by a third party who is independent from the investigators enrolling participants. After participants' eligibility is determined, the investigator will inform the third party (e.g., by electronic communication, telephone, or enter participant's information into the system if web-based e.g. REDCap, or through Interactive Voice Response System) to perform the allocation.

Explicitly stated using sequentially numbered opaque sealed envelopes with evidence that they were opened sequentially: Concealment using envelopes is susceptible to manipulation.[11] Systematic reviewers can evaluate trials using envelopes as definitely concealed only when sufficient details support rigor of the approach. First, trials need to explicitly state that the envelopes were sequentially numbered, opaque and sealed. In addition, there is evidence that the envelopes were opened sequentially only after the envelope has been irreversibly assigned to the participant (e.g., participant's details were written on the outside of the envelope and transferred to the assignment card by carbon or pressure-sensitive paper inside the envelope).[12]

Example:

1. A trial stated "Copenhagen Trial Unit is responsible for centralized and web-based 1:1 randomization according to a computer-generated allocation sequence list".[4 13] This trial used central allocation ("centralized and web-based").
2. A trial stated "After providing informed consent, the participants were randomly allocated in a 2:1 ratio to either an LCHF or a HCLF diet using computer-generated (REDCap) random allocation sequence with permuted blocks of 6 and 9 stratified by sex and the number of antidiabetic drugs (<2 versus ≥2) to balance the groups according to disease severity and to avoid potential gender differences". REDCap includes a randomization module that helps trialists implementing the allocation process and achieving allocation concealment.[14]

Probably Yes/Low

- **Trial explicitly stated using sequentially numbered, opaque, sealed envelopes without further details.**

  Explanation: *Without further details* means no evidence supporting the envelopes were opened sequentially only after the envelope has been irreversibly assigned to the participant (e.g., participant's details were written on the outside of the envelope and transferred to the assignment card by carbon or pressure-sensitive paper inside the envelope).

  Example: "Treatment regimens were included in opaque sealed numbered envelopes and the

envelope with the lowest number was always used for the consecutive patient".[15]

- **Drug trial in which participants and healthcare providers were blinded, with no further information on the allocation concealment method OR stated using envelopes but it remains unclear whether the envelopes were sequentially numbered opaque and sealed OR stated using drug containers but it remains unclear whether the drug containers were sequentially numbered and sealed.**
  Explanation: Typically, in a drug trial in which participants and healthcare providers are blinded, the medication is prepared by the pharmacy and leaves the pharmacy in a container that specifies this is for a particular participant without any further designation; thus, it is virtually impossible that the allocation is not concealed. To meet this criterion, judgement has to be that the trials are definitely or probably blinded (blinding of participants and blinding of healthcare providers items are evaluated as 'Definitely Yes' or 'Probably Yes').
  Example: A trial reported as double-blind (participants and healthcare providers were probably blinded) compared pantoprazole or placebo for stress ulcer prophylaxis, with no further information on the allocation concealment method.[16]

In addition:

1. If a trial explicitly stated it is concealed, but did not report any detail about allocation concealment method, systematic reviewers should assess this item as 'Probably Yes' (Step 1) or 'Probably Low' (Step 2).
2. If a trial used a minimization approach to allocating participants, participants and the individual investigator who enrolls the participants are unlikely to know how the multiple prognostic factors are accruing, thus systematic reviewers should assess this item as 'Probably Yes' (Step 1) or 'Probably Low' (Step 2).

Probably No/High

Unblinded drug trial or non-drug trial, with no further information on the allocation concealment method OR stated using envelopes but it remains unclear whether the envelopes were sequentially numbered opaque and sealed OR stated using drug containers but it remains unclear whether the drug containers were sequentially numbered and sealed.

Example:

1. An unblinded trial compared stress ulcer prophylaxis with lansoprazole OD 30 mg once daily versus no prophylaxis, with no further information on the allocation concealment method.[6]
2. An open-label trial stated "randomization was performed by the research nurse by using previously prepared closed and opaque envelopes". It is unclear whether the envelopes were sequentially numbered.[17]

Definitely No/High

- **Trial used an open random allocation schedule.**
  Explanation: *Open random allocation schedule* means trialists did not attempt to conceal allocation or there is clear evidence that the concealment method was compromised. Examples include: the allocation schedule was posted on a bulletin board; it is clearly that the same investigator generated the sequence and enrolled and assigned participants; drug containers were clearly not sequentially numbered, not sealed, or not of identical appearance or weight; envelopes were clearly not sequentially numbered, not opaque or sealed, or not opened sequentially.
- **Trial used a non-randomized allocation sequence that would be recognized as "quasi-**

randomization".

Explanation: *"Quasi-randomization"* cannot generate an unpredictable allocation sequence. Examples include: allocation based on dates of birth or admission; patient's hospital record number; alteration or rotation; allocation decided by clinicians or participants; allocation based on the results of a laboratory test; allocation by availability of the intervention.

Example: "Patients were randomized to each respective study arm by date of admission. Treatment regimens alternated every other month. Patients admitted into the unit during odd-numbered months received famotidine 20 mg IV every 12 h; patients admitted during even numbered months received lansoprazole suspension 30 mg suspended in 10 ml of an 8.4% sodium bicarbonate solution or apple juice, and administered via NG tube daily."[10]

Item 3 Blinding of participants

Step 1: Were participants blinded

Definitely Yes

Trial explicitly stated that participants were blinded.

Explanation: Systematic reviewers need to identify who are the *participants*. Most trials include patients as participants (they may explicitly state patients were blinded), but in some trials, participants are healthy people.

Example: A trial stated "patients, families, clinicians, and research staff were blinded."[18]

Probably Yes

No explicit statement about blinding of participants, and:

- **it is a placebo-controlled drug trial; or**
- **it is an active control drug trial (A vs. B) and mention of "double dummy" or that medications were identical or matched; or**
- **trial was described as "single blinded", "double blinded" or "triple blinded", and the best judgment is that one of the blinded groups is the participants; or**
- **participants not capable of distinguishing if they are receiving active or control intervention (e.g., neonates, severely demented).**

Example: A trial, although with no explicit statement about blinding of participants, stated "we conducted a prospective randomized double-blind parallel-group study. Study participants were randomly assigned to receive pantoprazole (40 mg in 10 mL of 0.9% saline IV) or placebo (10 mL of 0.9% saline IV)".[16] Here, double-blind is ambiguous as to who was blinded, but this is a placebo-controlled drug trial and it seems extraordinarily unlikely that patients could distinguish between a bag of IV fluid that does or does not contain pantoprazole.

Probably No

No explicit statement about blinding of participants, and:

- **it is an active control drug trial (A vs. B) and no mention of "double dummy" or that medications were identical or matched; or**
- **it is a non-drug trial; or**
- **trial was described as "single blinded" and the best judgement is that the single blinded group is someone other than the participants.**

Example: A randomized trial compared the effect of tidal peritoneal dialysis versus continuous renal replacement therapy on critically ill patients with acute kidney injury.[19] Since there is no explicit statement about blinding of participants and this is a non-drug trial in which it would appear to be impossible to blind patients to peritoneal dialysis or continuous renal replacement therapy, participants were probably not blinded.

Definitely No

- **Trial explicitly stated that participants were not blinded.**
- **Trial was described as "open-label" or "unblinded".**

Example:

1. A trial stated "Face-to-face treatment meant it was not possible to blind participants or the physical therapists delivering the interventions".[20]
2. A trial stated "This was a prospective, randomized, open-label, multicenter study".[17] "Open

label" constitutes an explicit statement of no blinding.

Step 2: Judge risk of bias related to blinding of participants

Issues to consider:

i) Were participants blinded (step 1)

ii) If unblinded, how likely unblinding of participants has influenced the outcome

- **How likely are participants expectations regarding effect of intervention to have influenced the outcome**

  Explanation:

  Unblinding of participants may cause the participants in intervention and control groups to have different expectations regarding the effect of the intervention they received and preconceptions about their outcomes, which may affect their actual outcome.[21]

  To judge how likely are participants expectations regarding effect of intervention to have influenced participants' outcome, systematic reviewers should consider two issues. First, how likely are unblinding of participants in intervention and control groups to have different expectations regarding the effect of the intervention they received. This is more likely in trials comparing an active intervention versus an inactive control than in trials comparing two active interventions in which participants are unlikely to have expectations regarding which intervention is superior. Second, how likely is the outcome to be influenced by participants' expectations regarding effect of intervention – this is decided by the type of outcome.

- **How likely are participant-initiated co-interventions to have influenced the outcome**

  Explanation:

  Unblinding of participants may lead to differential participant-initiated co-interventions between groups, thus influencing participants' outcome.

  Participant-initiated c*o-interventions* means any additional interventions that could potentially influence the outcome of interest that can be initiated by participants.

  To judge how likely are participant-initiated co-interventions to have influenced participants' outcome in a trial, systematic reviewers should consider two issues. First is the comparator in the trial. Trials comparing an active intervention versus an inactive control are more likely to have differential participant-initiated co-interventions than trials comparing two active interventions (participants in control group are more likely to seek other treatments when they know they receive an ineffective intervention or no treatment). Second, how easy was it for the participants to obtain co-interventions that had an appreciable impact on the outcome.

Definitely Low

- **Participants were definitely blinded.**

  Explanation: Trial explicitly stated that participants were blinded (systematic reviewers need to identify who are the participants).

  Example: A trial stated "patients, families, clinicians, and research staff were blinded."[18]

OR

- **Unblinding of participants very unlikely to have influenced the outcome because very**

unlikely participants expectations regarding effect of intervention have influenced the outcome and very unlikely participant-initiated co-interventions have influenced the outcome.

Example: An open randomized trial compared intrathoracic versus cervical anastomosis after totally or hybrid minimally invasive esophagectomy for esophageal cancer.[22] One outcome is in-hospital mortality. Although patients were unblinded (open-label), patients receiving intrathoracic and patients receiving cervical anastomosis were very unlikely to have different expectations regarding the effect of the intervention. Moreover, even if they did have different expectations, these were very unlikely to have influenced mortality. Finally, it is very likely there is no patient-initiated co-intervention that could influence mortality. Thus, for the outcome in-hospital mortality, there is 'Definitely Low' risk of bias related to blinding of participants.

Probably Low

- **Participants were probably blinded.**

  Explanation:

Probably Yes in Step 1	No explicit statement about blinding of participants, and: - it is a placebo-controlled drug trial; or - it is an active control drug trial (A vs. B) and mention of "double dummy" or that medications were identical or matched; or - trial was described as "single blinded", "double blinded" or "triple blinded", and the best judgment is that one of the blinded groups is the participants; or - participants not capable of distinguishing if they are receiving active or control intervention (e.g., neonates, severely demented).

Example: A trial, although with no explicit statement about blinding of participants, stated "we conducted a prospective randomized double-blind parallel-group study. Study participants were randomly assigned to receive pantoprazole (40 mg in 10 mL of 0.9% saline IV) or placebo (10 mL of 0.9% saline IV)".[16] Here, double-blind is ambiguous as to who was blinded, but this is a placebo-controlled drug trial and it seems extraordinarily unlikely that patients could distinguish between a bag of IV fluid that does or does not contain pantoprazole.

OR

- **Unblinding of participants unlikely to have influenced the outcome because unlikely participants expectations regarding effect of intervention have influenced the outcome and unlikely participant-initiated co-interventions have influenced the outcome.**

  Example: An open randomized trial compared intrathoracic versus cervical anastomosis after totally or hybrid minimally invasive esophagectomy for esophageal cancer.[22] One outcome is length of hospital stay. Although patients were unblinded, it is unlikely that patient's expectation regarding effect of intervention or patient-initiated co-interventions have influenced length of hospital stay. Thus, for the outcome length of hospital stay, there is 'Probably Low' risk of bias related to blinding of participants.

Probably High

Participants were definitely or probably not blinded, AND unblinding of participants likely to have influenced the outcome because participants expectations regarding effect of intervention likely to have influenced the outcome or participant-initiated co-interventions likely to have influenced the outcome.

Explanation:

Probably No in Step 1	No explicit statement about blinding of participants, and: - it is an active control drug trial (A vs. B) and no mention of "double dummy" or that medications were identical or matched; or - it is a non-drug trial; or - trial was described as "single blinded" and the best judgement is that the single blinded group is someone other than the participants.
Definitely No in Step 1	• Trial explicitly stated that participants were not blinded. • Trial was described as "open-label" or "unblinded".

Example: In a randomized, open-label trial, investigators randomly allocated 80 adult patients with acute low back pain into two groups: 40 patients received ibuprofen 400 mg three times daily for three days; 40 patients received a fixed-dose combination tablet of ibuprofen 200 mg plus paracetamol 325 mg three times daily for three days.[23] This is an open-label trial so patients were definitely not blinded. Because one group received a single drug but another received combination therapy, patients in intervention and control groups were likely to have different expectations regarding the efficacy of intervention. Thus, for the outcome pain intensity, it could be assessed as 'Probably High' risk of bias related to blinding of participants.

Definitely High

Participants were definitely or probably not blinded, AND unblinding of participants very likely to have influenced the outcome through participants expectations regarding effect of intervention or through participant-initiated co-interventions.

Explanation:

Probably No in Step 1	No explicit statement about blinding of participants, and: - it is an active control drug trial (A vs. B) and no mention of "double dummy" or that medications were identical or matched; or - it is a non-drug trial; or - trial was described as "single blinded" and the best judgement is that the single blinded group is someone other than the participants.
Definitely No in Step 1	• Trial explicitly stated that participants were not blinded; • Trial was described as "open-label" or "unblinded".

Example: A randomized, open-label trial compared melatonin and standard care versus standard care alone in hospitalized patients with COVID-19.[24] The primary outcome was sleep quality. This was an open-label trial, so patients were definitely not blinded. Sleep quality is likely to be influenced by patients' expectation regarding effect of intervention (patients receiving melatonin probably think melatonin could improve sleep quality and this belief may improve their sleep quality) or patient-initiated co-interventions (patients in control group were more likely to use hypnotics by themselves). Thus, for the outcome sleep quality, it could be assessed as 'Definitely High' risk of bias related to blinding of participants.

Item 4 Blinding of healthcare providers

Step 1: Were healthcare providers blinded

Definitely Yes

Trial explicitly stated that healthcare providers were blinded.

Explanation: *Healthcare providers* refer to the people who provide care and administer interventions. Some trials may not use the word "healthcare providers" explicitly, so systematic reviewers need to identify who were the healthcare providers in the trial. Most commonly they are physicians, nurses or members of other allied health professions.

Example: A trial stated "Physicians, bedside nurses and clinical pharmacists, other healthcare personnel, investigators, adjudicators, and the data analyst were blinded".[18]

Probably Yes

No explicit statement about blinding of healthcare providers, and:

- **it is a placebo-controlled drug trial; or**
- **it is an active control drug trial (A vs. B) and mention of "double dummy" or that medications were identical or matched; or**
- **trial was described as "single blinded", "double blinded" or "triple blinded", and the best judgment is that one of the blinded groups is the healthcare providers.**

Example: A trial, although with no explicit statement about blinding of healthcare providers, stated "we conducted a prospective randomized double-blind parallel-group study. Study participants were randomly assigned to receive pantoprazole (40 mg in 10 mL of 0.9% saline IV) or placebo (10 mL of 0.9% saline IV)".[16] Here, double-blind is ambiguous as to who was blinded, but this is a placebo-controlled drug trial and it seems extraordinarily unlikely that clinicians could distinguish between a bag of IV fluid that does or does not contain pantoprazole.

Probably No

No explicit statement about blinding of healthcare providers, and:

- **it is an active control drug trial (A vs. B) but no mention of "double dummy" or that medications were identical or matched; or**
- **it is a non-drug trial; or**
- **trial was described as "single blinded" and the best judgement is that the single blinded group is someone other than the healthcare providers.**

Example: A randomized trial compared the effect of tidal peritoneal dialysis versus continuous renal replacement therapy on critically ill patients with acute kidney injury.[19] Since there is no explicit statement about blinding of healthcare providers and this is a non-drug trial in which providers could easily distinguish whether patients are receiving peritoneal dialysis or continuous renal replacement therapy, healthcare providers were probably not blinded.

Definitely No

- **Trial explicitly stated that healthcare providers were not blinded.**
- **Trial was described as "open-label" or "unblinded".**

Example:

1. A trial stated "Face-to-face treatment meant it was not possible to blind participants or the physical therapists delivering the interventions".[20]
2. A trial stated "This was a prospective, randomized, open-label, multicenter study".[17] "Open

label" constitutes an explicit statement of no blinding.

Step 2: Judge risk of bias related to blinding of healthcare providers

Issues to consider:

i) Were healthcare providers blinded (step 1)

ii) If unblinded, how likely unblinding of healthcare providers has influenced the outcome

- **How likely healthcare provider-initiated co-interventions have influenced the outcome**

Definitely Low

Healthcare providers were definitely blinded.

Example: A trial stated "Physicians, bedside nurses and clinical pharmacists, other healthcare personnel, investigators, adjudicators, and the data analyst were blinded".[18]

Probably Low

- **Healthcare providers were probably blinded.**

  Explanation:

Probably Yes in Step 1	No explicit statement about blinding of healthcare providers, and: - it is a placebo-controlled drug trial; or - it is an active control drug trial (A vs. B) and mention of "double dummy" or that medications were identical or matched; or - trial was described as "single blinded", "double blinded" or "triple blinded", and the best judgment is that one of the blinded groups is the healthcare providers.

Example: A trial, although with no explicit statement about blinding of healthcare providers, stated "we conducted a prospective randomized double-blind parallel-group study. Study participants were randomly assigned to receive pantoprazole (40 mg in 10 mL of 0.9% saline IV) or placebo (10 mL of 0.9% saline IV)".[16] Here, double-blind is ambiguous as to who was blinded, but this is a placebo-controlled drug trial and it seems extraordinarily unlikely that clinicians could distinguish between a bag of IV fluid that does or does not contain pantoprazole.

OR

- **Unblinding of healthcare providers unlikely to have influenced the outcome because:**
  - **Unlikely there is any healthcare provider-initiated co-intervention that could potentially influence the outcome; or**
  - **Investigators have documented all healthcare provider-initiated co-interventions that could potentially influence the outcome and demonstrated similarity in use of all these co-interventions between groups.**

  Explanation: *Healthcare provider-initiated co-intervention that could potentially influence the outcome* means any additional intervention that could potentially influence the outcome of interest that can be initiated by healthcare providers, e.g., drug, advice, care, patient-healthcare provider interaction.[25]

  Example: Meningococcal serogroups A, B, C, W, and Y cause nearly all meningococcal disease. In a trial, investigators randomly allocated healthy individuals (age 10-25 years) to one group

receiving 0·5 mL of a MenABCWY vaccine (months 0 and 6) and placebo (month 0), or another group receiving MenB-FHbp vaccine (months 0 and 6) and MenACWY-CRM vaccine (month 0) via intramuscular injection into the upper deltoid.[26] Investigators compared the proportion of participants who achieved at least a four-fold increase in hSBA (serum bactericidal antibody using human complement) titers from baseline for each serogroup. Although the study staff administering the vaccine were unblinded, it is unlikely there are any healthcare provider-initiated co-interventions that could potentially influence the outcome, the risk of bias is 'Probably Low'.

Probably High

Healthcare providers were definitely or probably not blinded, AND unblinding of healthcare providers likely to have influenced the outcome because there are healthcare provider-initiated co-interventions that could potentially influence the outcome.

Explanation:

Probably No in Step 1	No explicit statement about blinding of healthcare providers, and: - it is an active control drug trial (A vs. B) but no mention of "double dummy" or that medications were identical or matched; or - it is a non-drug trial; or - trial was described as "single blinded" and the best judgement is that the single blinded group is someone other than the healthcare providers.
Definitely No in Step 1	• Trial explicitly stated that healthcare providers were not blinded. • Trial was described as "open-label" or "unblinded".

Example: A randomized, open-label trial compared remdesivir and standard care versus standard care alone for the treatment of patients in hospital with COVID-19.[27] One outcome was duration of hospital stay. This is an open-label trial so healthcare providers were definitely not blinded. Unblinding of healthcare providers may possibly have influenced duration of hospital stay because duration of hospital stay is to a large extent under the control of the clinicians.

Definitely High

Healthcare providers were definitely or probably not blinded, AND unblinding of healthcare providers very likely to have influenced the outcome because there are healthcare provider-initiated co-interventions that could potentially influence the outcome and investigators have documented dissimilarity in any of these co-interventions between groups.

Explanation:

Probably No in Step 1	No explicit statement about blinding of healthcare providers, and: - it is an active control drug trial (A vs. B) but no mention of "double dummy" or that medications were identical or matched; or - it is a non-drug trial; or - trial was described as "single blinded" and the best judgement is that the single blinded group is someone other than the healthcare providers.
Definitely No in Step 1	• Trial explicitly stated that healthcare providers were not blinded. • Trial was described as "open-label" or "unblinded".

Example: A randomized trial compared the effectiveness of Lopinavir/ Ritonavir/ Hydroxychloroquine (KH) versus Atazanavir/ Ritonavir/ Dolutegravir/ Hydroxychloroquine (ADH) in COVID-19 patients.[28] Healthcare providers were probably not blinded. Corticosteroid was an

important healthcare provider-initiated co-intervention that could influence the outcome mortality. The mortality rate in ADH group (3/32) was lower than KH group (6/30); while the proportion of corticosteroid administration in the ADH group (9/32) was higher than in the KH group (2/30) (p=0.03). The difference in corticosteroid was unlikely to be a function of the effect of interventions since the KH group with more mortality (worse outcome) received less corticosteroid. Thus, we conclude there is 'Definitely High' risk of bias related to blinding of healthcare providers.

Item 5 Blinding of outcome assessors

Step 1: Were outcome assessors blinded

Explanation: Outcome assessors can be different for different outcomes. Reviewers need to identify, for the outcome of interest, who are the outcome assessors. They could be participants (for participant-reported outcomes), healthcare providers (for healthcare provider-assessed outcomes), or independent assessors.

Definitely Yes

Trial explicitly stated that the outcome assessors or adjudicators (people making the measurement or assessment) were blinded.

Explanation: Reviewers should carefully consider who were the assessors or adjudicators for the outcome of interest, especially for trials with multiple outcomes. Consider, for instance, a trial with multiple outcomes explicitly stated "outcome assessors" were blinded; however, for a patient-reported outcome actually the outcome assessors were not blinded (because in this case patients were outcome assessors and patients were not blinded).

Example: A trial stated "Two clinicians blinded to allocation and to each other's assessments adjudicated all suspected clinical outcomes".[18]

Probably Yes

No explicit statement about blinding of outcome assessors or adjudicators, and:

- **it is a placebo-controlled drug trial; or**
- **it is an active control drug trial (A vs. B) and mention of "double dummy" or that medications were identical or matched; or**
- **trial was described as "single blinded", "double blinded" or "triple blinded", and the best judgement is that one of the blinded groups is the outcome assessors.**

Example: A trial, although with no explicit statement about blinding of outcome assessors, stated "we conducted a prospective randomized double-blind parallel-group study. Study participants were randomly assigned to receive pantoprazole (40 mg in 10 mL of 0.9% saline IV) or placebo (10 mL of 0.9% saline IV)".[16] For the outcome overt gastrointestinal bleeding, clinicians or nurses were the outcome assessors. Here, double-blind is ambiguous as to who was blinded, but this is a placebo-controlled drug trial and it seems extraordinarily unlikely that clinicians or nurses could distinguish between a bag of IV fluid that does or does not contain pantoprazole.

Probably No

No explicit statement about blinding of outcome assessors or adjudicators, and:

- **it is an active control drug trial (A vs. B) but no mention of "double dummy" or that medications were identical or matched; or**
- **it is a non-drug trial; or**
- **trial was described as "single blinded" and the best judgement is that the single blinded group is someone other than the outcome assessors.**

Example: A randomized trial compared the effect of tidal peritoneal dialysis versus continuous renal replacement therapy on critically ill patients with acute kidney injury.[19] For the outcome time to recovery of renal function, clinicians were the outcome assessors. Since there is no explicit statement about blinding of outcome assessors/clinicians and this is a non-drug trial (tidal peritoneal dialysis and continuous renal replacement therapy are easily distinguished), clinicians were probably not blinded.

Definitely No

- **Trial explicitly stated that the outcome assessors or adjudicators were not blinded.**
- **Trial was described as "open-label" or "unblinded".**

Example: "This was a prospective, randomized, open-label, multicenter study".[17]

Step 2: Judge risk of bias related to blinding of outcome assessors

Issues to consider:

i) Were outcome assessors (people making the measurement or assessment) blinded

Explanation: Outcome assessors can be different for different outcomes, thus evaluation of blinding of outcome assessors must consider the outcome. Reviewers need to identify, for the outcome of interest, who are the outcome assessors. They could be participants (for participant-reported outcomes), healthcare providers (for healthcare provider-assessed outcomes), or independent assessors.

ii) If unblinded, how likely unblinding of outcome assessors has influenced the outcome assessment

- **Degree of judgement/subjectivity involved in the outcome assessment (more judgment/subjectivity more likelihood of bias)**

Definitely Low

- **Outcome assessors were definitely blinded.**

  Explanation: Trial explicitly stated that the outcome assessors or adjudicators (people making the measurement or assessment) were blinded.

  Example: A trial stated "Two clinicians blinded to allocation and to each other's assessments adjudicated all suspected clinical outcomes".[18]

OR

- **Outcome is all-cause mortality.**

  Explanation: Assessment of all-cause mortality involves no judgement.

  Example: A randomized open-label trial compared the effect of high-dose hemodiafiltration versus continuation of high-flux hemodialysis on patients with kidney failure.[29] For the outcome all-cause mortality, there is 'Definitely Low' risk of bias.

Probably Low

- **Outcome assessors were probably blinded.**

  Explanation:

Probably Yes in Step 1	No explicit statement about blinding of outcome assessors or adjudicators, and: - it is a placebo-controlled drug trial; or - it is an active control drug trial (A vs. B) and mention of "double dummy" or that medications were identical or matched; or - trial was described as "single blinded", "double blinded" or "triple blinded", and the best judgement is that one of the blinded groups is the outcome assessors.

Example: A trial, although with no explicit statement about blinding of outcome assessors,

stated "we conducted a prospective randomized double-blind parallel-group study. Study participants were randomly assigned to receive pantoprazole (40 mg in 10 mL of 0.9% saline IV) or placebo (10 mL of 0.9% saline IV)".[16] For the outcome overt gastrointestinal bleeding, clinicians or nurses were the outcome assessors. Here, double-blind is ambiguous as to who was blinded, but this is a placebo-controlled drug trial and it seems extraordinarily unlikely that clinicians or nurses could distinguish between a bag of IV fluid that does or does not contain pantoprazole.

OR

- **Unblinding of outcome assessors likely could not have influenced the outcome assessment because the outcome assessment involves minimal judgement (other objective outcomes, e.g., laboratory measurement, hospital admission, mechanical ventilation).**
  Example: A randomized trial compared the effect of tidal peritoneal dialysis versus continuous renal replacement therapy on critically ill patients with acute kidney injury.[19] For the outcome length of ICU stay, outcome assessors were probably not blinded (no explicit statement about blinding of outcome assessors and this is a non-drug trial). However, since assessment of length of ICU stay involves minimal judgement, there is 'Probably Low' risk of bias.

Probably High

Outcome assessors were definitely or probably not blinded, AND unblinding of outcome assessors likely could have influenced the outcome assessment because the outcome assessment involves some judgment (e.g., cause-specific mortality).

Explanation:

Probably No in Step 1	No explicit statement about blinding of outcome assessors or adjudicators, and: - it is an active control drug trial (A vs. B) but no mention of "double dummy" or that medications were identical or matched; or - it is a non-drug trial; or - trial was described as "single blinded" and the best judgement is that the single blinded group is someone other than the outcome assessors.
Definitely No in Step 1	• Trial explicitly stated that the outcome assessors or adjudicators were not blinded. • Trial was described as "open-label" or "unblinded".

Example: A randomized open-label trial compared the effect of high-dose hemofiltration versus continuation of high-flux hemodialysis on patients with kidney failure.[29] For the outcome death from cardiovascular causes, clinicians were the outcome assessors. Since outcome assessors were not blinded (open-label) and assessment of death from cardiovascular causes involves some judgement, there is 'Probably High' risk of bias.

Definitely High

Outcome assessors were definitely or probably not blinded, AND unblinding of outcome assessors could have influenced the outcome assessment because the outcome assessment involves considerable judgment by participant or adjudicator (e.g., symptoms and symptom scores, quality of life, seizure occurrence).

Explanation:

Probably No in Step 1	No explicit statement about blinding of outcome assessors or adjudicators, and: - it is an active control drug trial (A vs. B) but no mention of "double dummy" or

	that medications were identical or matched; or - it is a non-drug trial; or - trial was described as "single blinded" and the best judgement is that the single blinded group is someone other than the outcome assessors.
Definitely No in Step 1	• Trial explicitly stated that the outcome assessors or adjudicators were not blinded. • Trial was described as "open-label" or "unblinded".

Example: A randomized open-label trial compared the efficacy of ibuprofen plus paracetamol versus ibuprofen alone on patients with acute low back pain.[23] For the outcome pain intensity assessed using a visual analogue scale, patients were outcome assessors. Since patients were not blinded (open-label) and assessment of pain intensity involves considerable judgement, there is 'Definitely High' risk of bias.

Item 6 Outcome data not included in analysis

Two approaches for assessing risk of bias related to item 6

Item 6 addresses the randomized participants whose outcome data were not included in the analysis; that is, the participants whose outcome data were unavailable to systematic reviewers. Outcome data unavailable to reviewers could be a result of outcome data unavailable to trialists (e.g. lost to follow-up), or due to trialists excluded the participants for whom outcome data were available from the analysis because the participants did not adhere to the protocol (i.e., per-protocol analysis).

Rationale for completing only step 1
Step 1 asks the systematic reviewers to extract the number of participants whose outcome data were not included in analysis. The systematic review teams can use these numbers to assess risk of bias associated with missing data across the entire body of evidence at the meta-analysis level. The process of doing so begins with a complete-case analysis followed by an analysis imputing data for participants in each trial who were not included in the analysis.[1 2]

The review team can always use this approach either for binary or continuous outcomes but,[2] because the information needs is seldom available, rarely for time-to-event outcomes.[1]

If systematic review teams decide to use this approach to assess risk of bias associated with this item at the meta-analysis level, they can ask the systematic reviewers assessing individual trials to complete only step 1.

Rationale for completing both steps
Step 1 can serve to inform step 2. Step 1 asks systematic reviewers to extract the number of participants not included in analysis and calculate the overall percentage of participants not included in analysis. This percentage provides the information required for step 2, which asks systematic reviewers to classify trials into definitely low, probably low, probably high and definitely high risk of bias.

Set thresholds for step 2

If systematic reviewers need to complete both steps, systematic review teams need to set a threshold - the missing percentage - for each response option, as example below.

Step 2: Judge risk of bias related to the overall percentage of participants not included in analysis

Issue to consider: Was the overall percentage of participants not included in analysis acceptably low

Note: Systematic review teams need to set and fill in the threshold for each response option. See manual for instructions and example thresholds.

☐ **Definitely Low**	Percentage of participants not included in analysis is < 5 %
☐ **Probably Low**	• Percentage of participants not included in analysis is 5 % to < 10 % • If the trial did not mention whether there were participants not included in analysis, a substantial loss to follow-up is unlikely (e.g., ICU mortality)
☐ **Probably High**	• Percentage of participants not included in analysis is 10 % to < 15 % • If the trial did not mention whether there were participants not included in analysis, a substantial loss to follow-up is likely (e.g., 1-year quality of life)
☐ **Definitely High**	Percentage of participants not included in analysis is ≥ 15 %
Support for judgement:	

Specify post-randomization exclusion without bias (optional)

Review teams aiming at the highest level of rigor will consider one additional issue in addressing item 6. Usually, 'N total' is the number of randomized participants. However, sometimes, post-randomization exclusion does not introduce bias ('N exclusion without bias'). In this case, 'N excluded without bias' should be excluded from the 'N total' and 'N not analyzed'. Review teams will find a full explanation of the issue of exclusion without bias in:
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1124168/[30]

Manual for completing step 1

'N total'
Usually, 'N total' is the number of randomized participants.

However, sometimes, post-randomization exclusion may be legitimate and does not introduce bias ('N exclusion without bias'). In this case, 'N exclusion without bias' should be excluded from the 'N total' and 'N not analyzed'.

Systematic reviewers should refer to the instructions that the systematic review team provided regarding in which cases post-randomization exclusion does not introduce bias.

'N not analyzed'
'N not analyzed' is: among the 'N total', how many participants' outcome data were not included in the analysis of the outcome of interest. Analyses of different outcomes may include different numbers of participants.

There are two main reasons for not including in analysis:

1) Missing outcome data

   Participants outcome data were unavailable to the trialists because of loss to follow-up, outcome data cannot be collected because of withdrawal of consent, etc.

   For time-to-event outcomes, missing outcome data refers to the participants who are censored because of missing follow-up data (i.e., informative censoring) rather than because of end of study.[1] Thus, systematic reviewers should count the number of participants who were censored because of missing follow-up data in 'N not analyzed'.

2) Per-protocol analysis

   Trialists may exclude the participants for whom outcome data were available from the analysis because the participants did not adhere to the protocol. For example, excluding because the participants did not receive assigned intervention, not meeting inclusion criteria, etc.

   Per-protocol analysis is a type of deviation from intention-to-treat analysis. Intention-to-treat analysis means that all participants for whom outcome data were available were included in the analysis and analyzed in the group to which they were randomized. Another type of deviation from intention-to-treat analysis is as-treated analysis, which is addressed as an optional item rather than in this item.

If the systematic review team specified in which cases post-randomization exclusion does not introduce bias, these cases should not be counted in 'N not analyzed'.

<u>Example:</u>
In a trial with 323 randomized patients, 35 were subsequently excluded from analysis: 1 patient died suddenly within 2 hours after randomization, 18 underwent mechanical ventilation <48h (did not meet the inclusion criterion of mechanical ventilation >48h), and 16 were not assessable because of missing important data.[15] The outcome of interest is clinically significant stress-related upper gastrointestinal bleeding.

Consider, if the systematic review team provides instructions that when considering non-death

outcomes post-randomization exclusion because of deaths does not introduce bias, then for the combined group 'N total' = 'N randomized' (323) - 'N exclusion without bias' (1 patient who died) = 322. Among the 322 patients, 34 were not included in analysis, thus 'N not analyzed' is 34. Thus, overall '% not analyzed' = 34/322 = 10.6%

Manual for completing step 2

In step 1, systematic reviewers should have calculated the overall percentage of participants not included in analysis. In step 2, systematic reviewers should, based on the thresholds that set by the systematic review team, classify the trial into ''Definitely Low', 'Probably Low', 'Probably High' or 'Definitely High'.

If the trial did not mention whether there were participants not included in analysis, classify the trial as 'Probably Low' or 'Probably High' depending on the nature of the outcome (easy to follow such as death, versus difficult to follow such as quality of life) and the follow-up time (short such as 1 week versus long such as two years).

Example:
A randomized trial with only an abstract compared pantoprazole with famotidine for stress ulcer prevention in ICU.[31] In the results section it stated: "Patients were randomized to IV pantoprazole (n=68) and IV famotidine (n=61) … Stress-related mucosal bleeding occurred in four patients in the pantoprazole group (3 overt bleeds, p=0.36 and 1 CSUGB, p=0.34) and one patient in the famotidine group (overt bleed). Patients given IV pantoprazole had similar mean ICU/CCU LOS (6.7 vs. 6.5 days, p=0.37) and duration on mechanical ventilation (6.5 vs. 6.3 days, p=0.39). Patients on pantoprazole (n=22) experienced more adverse effects than famotidine (n=10); hypomagnesemia (13 vs. 5) and nausea/vomiting (7 vs. 3); C. difficile diarrhea was the same in both groups (2 vs. 2)." This trial is only available as abstract and there is no detail regarding whether there were patients not included in analysis. Since the patients were followed until transferred out of the ICU (follow-up time is short), the trial could be classified as 'Probably Low'.

In addition:
If participants were not included in the trial analysis, but they were followed successfully and their outcome data were reported in trial publication or provided after inquiry, systematic reviewers should include these outcome data in their own analysis (i.e., conduct the intention-to-treat analysis themselves). In this case, these participants should not be counted as "N not analyzed" in this item.

Example: A randomized trial compared pantoprazole and enteral nutrition versus placebo and enteral nutrition for stress ulcer prophylaxis in critically ill patients with mechanical ventilation.[32] Investigators randomly allocated 124 patients. Of the 62 patients in pantoprazole group, 7 were excluded because they were extubated within the first 24h (one inclusion criterion was mechanical ventilation >48h; thus these patients did not adhere to the protocol); thus, the trial analysis included 55 patients. Of the 62 in placebo group, 15 were excluded because they were extubated within the first 24h; thus, the trial analysis included 47 patients. However, since the trial stated that none of these 22 patients developed any primary or secondary outcomes, these patients can be added back into the analysis. For instance, the trial reported one patient in each group (1/55 in

pantoprazole, 1/47 in placebo) developed overt bleeding. Systematic reviewers should extract the data for bleeding as 1/62 in pantoprazole group and 1/62 in pantoprazole group. In this case, all patients were included and analyzed in the group to which they were randomized.

Decide which if any optional items to include

In addition to the six core items (item 1 to 6), we provide eight optional items. Systematic review teams should decide whether to bring them to the attention of the systematic reviewers who assess individual trials (i.e., risk of bias assessors).

We include these eight items as optional rather than core items mainly either because the information required to make judgements is not commonly reported or because problems with these items occurs infrequently, or both.

However, these items may still be worth considering in certain circumstances.

Table 3 presents these eight optional items, reasons why they are not justified as core items, reasons why we include them as optional items, and further considerations regarding reasons systematic reviewers might or might not include these items in the ROBUST-RCT they use in their systematic review.

We strongly recommend that the systematic review teams consider each of the optional items carefully and judge whether they are relevant to their particular reviews.

If systematic review teams decide to include optional item(s), they need to provide the risk of bias assessors the instructions for assessing the items. Please see example instruction for the optional item 1 in Table 4.

Table 3. Optional items

	Reasons for not including as a core item	Reasons for including as an optional item in this instrument	Considerations regarding inclusion of this item in a systematic review
Optional item 1: **Whether baseline prognostic factors were balanced between groups**	• When considering this item, one has to consider whether a baseline characteristic is an important prognostic factor in the particular context. • We have already included random sequence generation and allocation concealment as core items. Prognostic imbalance due to inappropriate randomization will be at least in part covered by these core items. If randomization is conducted properly and sample size is sufficient, prognostic imbalance would happen rarely and be simply due to chance. • While prognostic imbalance will often happen in small studies, it is much less likely across the entire range of studies (prognostic imbalance in one study is likely to be ameliorated by distribution of prognostic variables in other studies). • Empirical evidence regarding this item actually creating bias is very uncertain.	• When sample size is small, investigators may generate sequence appropriately and conceal and still have imbalance of prognostic factors simply by chance. • When a baseline characteristic with known appreciable prognostic power is imbalanced between groups, it may create serious bias.	• If there is problem with random sequence generation or allocation concealment (thus high risk of bias related to sequence generation or concealment item), no need to include this item. • If random sequence generation and allocation concealment performed well, and there is an important imbalance in important prognostic factors, an extra risk of bias exists. This item captures this problem. • The larger the imbalance in a factor, the stronger the prognostic power of the factor, the larger the number of trials in which this exists and the larger the weight of these trials in the meta-analysis, the more likely one would include this item.
Optional item 2: **Whether co-interventions were balanced between groups in**	• It is difficult for non-expert systematic reviewers to judge what is a sufficiently important co-intervention and, if it is sufficiently important, when there is enough	• When sample size is small, investigators may generate sequence, conceal and conduct blinding appropriately and still have imbalance	• If there is problem with random sequence generation or allocation concealment (thus high risk of bias), or participants or healthcare providers

blinded trials	imbalance to consider as high risk of bias. • It is difficult to distinguish whether the imbalance in co-interventions is a risk of bias issue or is a function of the effect of intervention (e.g., participants experiencing adverse events receive additional drug to treat the adverse events). • If randomization is conducted properly and participants and healthcare providers are blinded, imbalance in co-interventions happens rarely and is simply due to chance.	of co-interventions simply by chance. • When an important co-intervention is imbalanced between groups and the imbalance in the co-intervention is not a function of the effect of intervention, it may create serious bias.	were unblinded, no need to include this item. • If randomization performed well and participants and healthcare providers were blinded (thus low risk of bias), and there is an important imbalance in important co-interventions, an extra risk of bias exists. This item captures this problem. • If the imbalance in co-intervention is a function of the effect of intervention, this is not a risk of bias problem. • The larger the imbalance in a co-intervention, the stronger the influence of the co-intervention on the outcome, the larger the number of trials in which this exists and the larger the weight of these trials in the meta-analysis, the more likely one would include this item.
Optional item 3: **Whether outcome assessment or data collection differed between groups**	• Information required to make judgements is not commonly reported. • Even if reported, systematic reviewers will not make the judgement easily. • Different outcome measurement or data collection between groups happens rarely.	• If trials do report the information, and there is a difference between groups in the way of outcome assessment or data collection, it could lead to risk of bias.	• The problem is much more likely to occur in unblinded situations, but it is theoretically possible it could occur in blinded situations as well.

	• No empirical evidence supports this item.		
Optional item 4: **Whether follow-up time, frequency, or intensity of outcome assessment differed between groups**	• Information required to make judgement is not commonly reported. • Even if reported, systematic reviewers cannot make the judgement easily. • Different follow-up time, frequency or intensity of outcome assessment between groups happens rarely. • No empirical evidence supports this item.	• If trials do report the information, and there is a difference between groups in the follow-up time, frequency or intensity of outcome assessment, it could lead to risk of bias.	• The problem is much more likely to occur in unblinded situations, but it is theoretically possible it could occur in blinded situations as well.
Optional item 5: **Whether outcome measurement method was valid (ie, validity of outcome measurement)**	• It is difficult for non-expert systematic reviewers to judge whether the outcome assessment method is valid or not. Even if the method has been validated, it may not be valid in the particular setting of the systematic review. • For many outcomes e.g., mortality, stroke, admission to hospital, this item is not applicable. • No empirical evidence supports this item.	If reviewers have major concerns that the outcome assessment method is not valid, risk of bias is present.	
Optional item 6: **When investigators conducted an as-treated analysis, was the percentage of participants not analyzed in the groups to which they were randomized sufficiently**	• Although “as treated analysis” (crossover occurred and participants were analyzed according to the intervention they received rather than the intervention they allocated to) is a serious problem, it happens very rarely.	• If in a randomized trial, crossover occurred and participants were analyzed according to the intervention they received rather than the intervention they allocated to (commonly referred to as “as-treated analysis”), and the percentage of participants analyzed in wrong group is large, there is serious risk of bias. • If in the context area of interest (e.g. surgical trial), participants were likely to switch to the other intervention group and trialists were likely to analyze these participants	

low	• No empirical evidence supports this item.	according to the intervention they received rather than the intervention they allocated to, systematic reviewers should include this item. • The larger the percentage of patients in all trials included in trials that used an as-treated analysis, and the larger the percentage who crossed in each of the trials, the stronger the case for including this item. • Question for this item could be: Among the participants who were included in analysis, was the percentage of participants analyzed in wrong group acceptably low.
Optional item 7: **Whether there was selective reporting**	• The protocol or analysis plan is not always available and if available it does not always provide sufficient detail to make the judgment regarding whether there are multiple choices regarding relevant analyses plans. • It is difficult for non-expert systematic reviewers to judge whether the selection is or is not based on the apparent magnitude of effect. • Systematic reviewers usually go back to the two-by-two tables and use the raw data to do their analysis, so the problem does not occur frequently. • Empirical evidence is very uncertain.	If the reported result is selected from multiple effect estimates that are available to the trialists (e.g., multiple outcome measurement instruments measuring the same construct, multiple time points, or multiple alternative analytic strategies) and the basis of the choice is the apparent magnitude of effect, or there is inconsistency between the protocol and the publication report, there may be a serious risk of bias.
Optional item 8: **Whether the trial was terminated early for benefit**	• Early termination for benefit is unlikely to be a risk of bias when a meta-analysis includes a large number of trials in which the contribution of the trials that stopped early for benefits is small.	When trials that terminated early for benefit have a substantial weight in the meta-analysis, there will be a serious risk of bias. For binary outcomes, if the trials had modest numbers of events, the problem will be more serious.[33] Systematic reviewers should be alert to this problem and certainly include it when the trials terminated early for benefit contribute substantial weight to the

	• Early termination does not occur frequently in some certain clinical areas.	pooled estimate.

Table 4. Example instruction for optional item 1

Optional item	Instruction	Judgement	Support for judgement
Whether the baseline prognostic factors were balanced between groups	**Definitely Yes/Low:** There was no imbalance in any important prognostic factors. **Probably Yes/Low:** • The imbalance in prognostic factor(s) may not result in important risk of bias because the imbalance may not be large enough and the prognostic power of the factor(s) may not be strong enough. • No problem with random sequence generation and allocation concealment, and the trial did not report prognostic factors. **Probably No/High:** The imbalance in prognostic factor(s) may result in important risk of bias because the imbalance may be large enough and the prognostic power of the factor(s) may be strong enough. **Definitely No/High:** The trial reported important imbalance in important prognostic factor(s). **Not applicable:** There is problem with random sequence generation or allocation concealment (definitely/probably high risk of bias related to sequence generation or concealment item), thus no need to assess this optional item.		

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